Legislative Coordination and Institutional Change

A Bipartite Network Analysis of Coalition Dynamics in the Brazilian Chamber of Deputies

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Slides & paper

The puzzle

- Polarization should make legislatures **ungovernable**.
- Brazil 2014–16: impeachment · corruption scandals · recession · polarization.
- Yet the Chamber **kept forming majorities**, across impeachment and a change of power.
- The literature's crisis typology: **collapse · fragmentation · realignment · institutionalization**. Brazil fits **none**.
- **How? And at what cost?**

Coalitional presidentialism, and a cluster of shocks

- Strong president, **no majority** → govern by **coalition** (Abranches 1988).
- Levers: **budget · posts · agenda**; the **Centrão** sells support as the president's *agent*.
- Recent work reads a **weakening presidency and a rising Congress**, much of it via **budget powers** once a pillar of the president's toolkit.
- 2014–16, all at once: impeachment · corruption scandals · recession · **and the president's budget lever removed**.
- **Shocks co-occur in a tight 2014–16 window** → no single lever separates cleanly from the rest (so the design weighs the alternatives one by one).

The structural question, and some rival explanations

- The open question is structural: **How did coordination reorganize after the 2014–2016 shocks? What happened to the pragmatic center, and at what cost?**
- **Then weigh the leading alternatives, one at a time:**
- Is it **turnover** (new members) or a one-off **crystallization**?
- Is it the **Cunha effect** (the agenda-setter) or the **fiscal channel**: does re-sorting concentrate on economic-fiscal rather than rights-and-moral bills?

The tests ask whether a durable pivot remains once turnover, agenda, and fiscal explanations are removed.

The methodological approach

- **Measure:** roll-calls → **deputy × bill** graph → **communities per legislature**.
- **Classify bills:** fiscal vs moral, **rule-based** + local **LLM cross-check** (Qwen 2.5 32B), human-validated ($n=294$, $\alpha=0.84$).
- **Compare:** across the 2014–16 break; a **within-deputy panel** separates behavior from turnover.
- **Weigh alternatives, not estimate:** near-collinear shocks → **assess each rival in turn**.
- **Stress-test:** Q* null, 12-cell threshold grid, alternative projections and algorithms.

Votes as a bipartite incidence matrix

$$X_{ij} = \begin{cases} +1 & \text{deputy } i \text{ voted } \textbf{Yes} \\ -1 & \text{voted } \textbf{No} \\ 0 & \text{abstention, obstruction, or absent} \end{cases}$$

Deputy × bill, before projecting to the agreement graph.

The coordination metric: agreement score

$$W_{ij} = \frac{|\{v : S_{iv} = S_{jv} \neq 0\}|}{|\{v : S_{iv} \neq 0 \text{ and } S_{jv} \neq 0\}|}, \qquad S = \operatorname{sign}(X)$$

where W_{ij} = agreement rate between deputies i and j • $S = \operatorname{sign}(X)$ = signed vote matrix ($+1$ Yes, -1 No, 0 otherwise) • v = a recorded vote

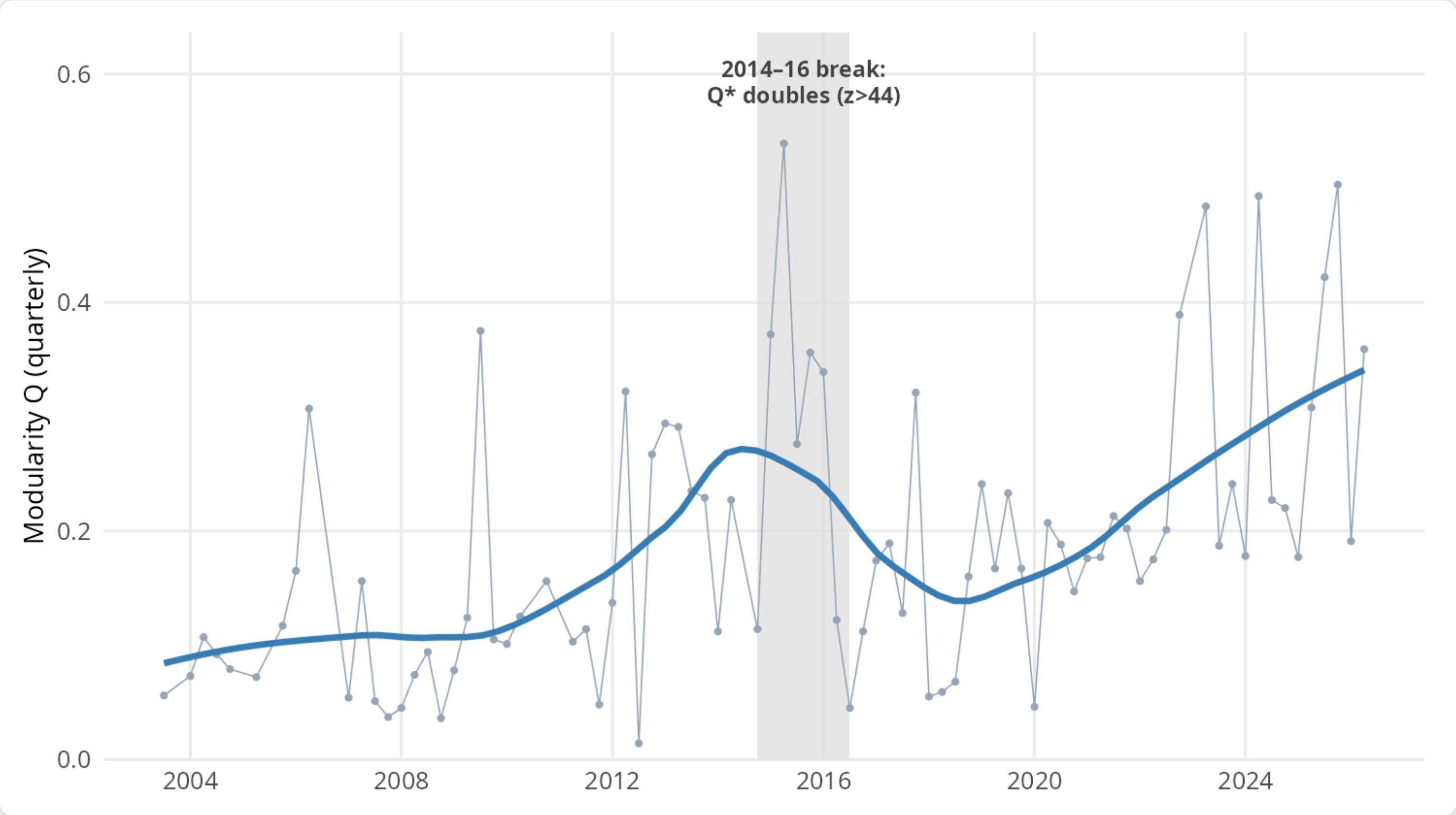
numerator = votes where i, j agree • denominator = votes both attended • edge set to 0 if fewer than 30 shared votes

Krehbiel (1993) co-voting agreement, normalised by joint attendance.

What the ERGM can and cannot establish

- **Bipartite pre-test** (deputy × bill): **did not mix** on the dense 54L network, so its coefficients are not interpreted.
- **Cross-sectional** (agreement graph, per legislature): **runs**. Net of party and state, shared **Centrão status becomes an organizing boundary** after the crisis.
- **Temporal, btergm** (agreement graph over time): **runs**. Recovers **closure** and **tie persistence**, but cannot isolate EC 86 (near-collinear shocks, VIF 11.6).
- **The binding constraint is identification, not model availability.**

A structural break: modularity spikes in 2015–16



Quarterly modularity rises from ≈.15 to .51 in 2016.

Null-corrected modularity, Q*

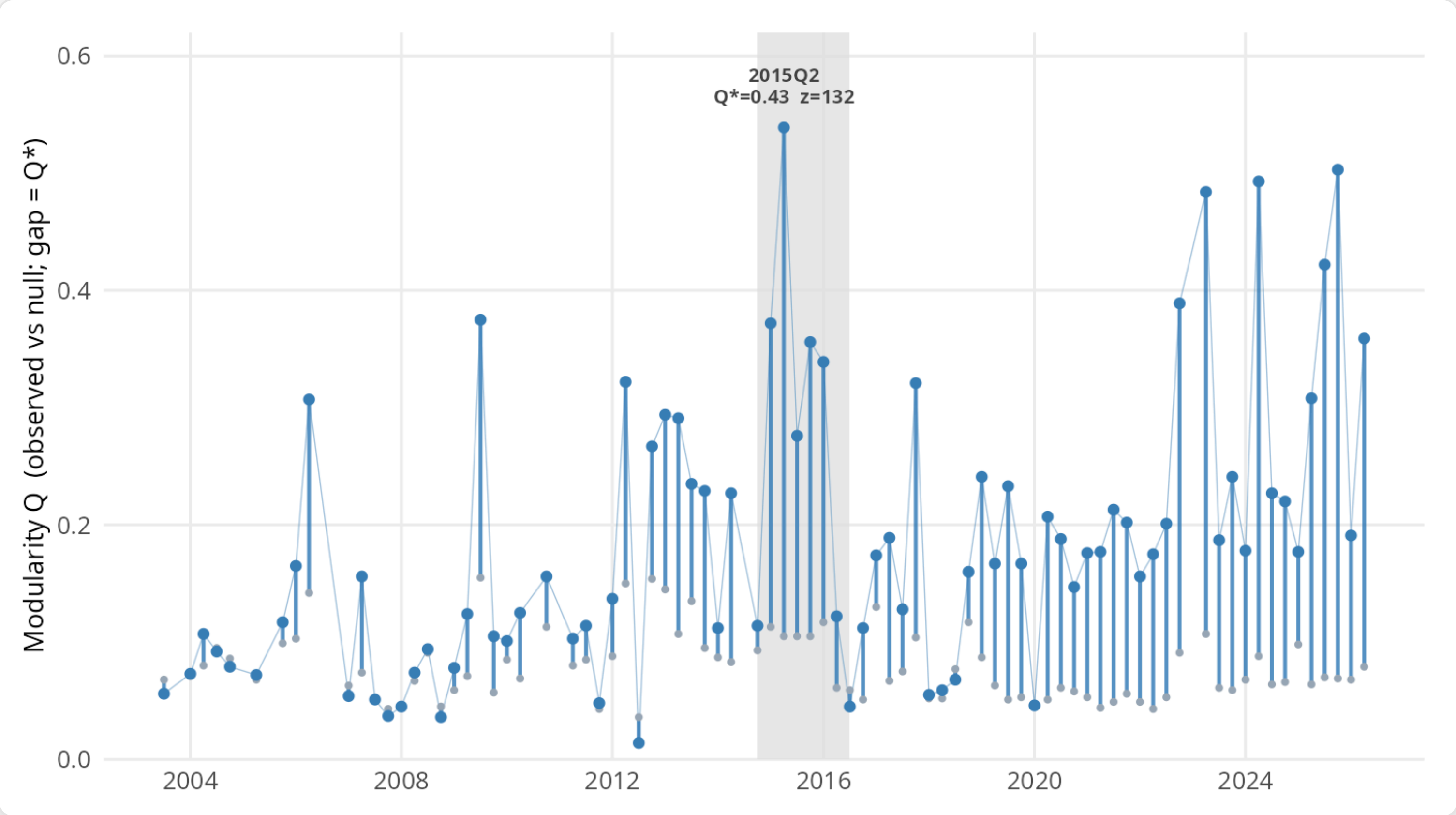
$$Q = \frac{1}{2m}\sum_{ij}\left(A_{ij} - \frac{k_i k_j}{2m}\right)\delta(c_i, c_j)$$

$$Q^* = Q_{\text{obs}} - \langle Q_{\text{null}}\rangle, \qquad z = \frac{Q_{\text{obs}} - \mu_{\text{null}}}{\sigma_{\text{null}}}$$

where A_{ij} = agreement-graph weight • k_i = weighted degree of i • m = total edge weight • $\delta(c_i, c_j) =$ 1 if i, j share a community, else 0 • null = within-vote permutations

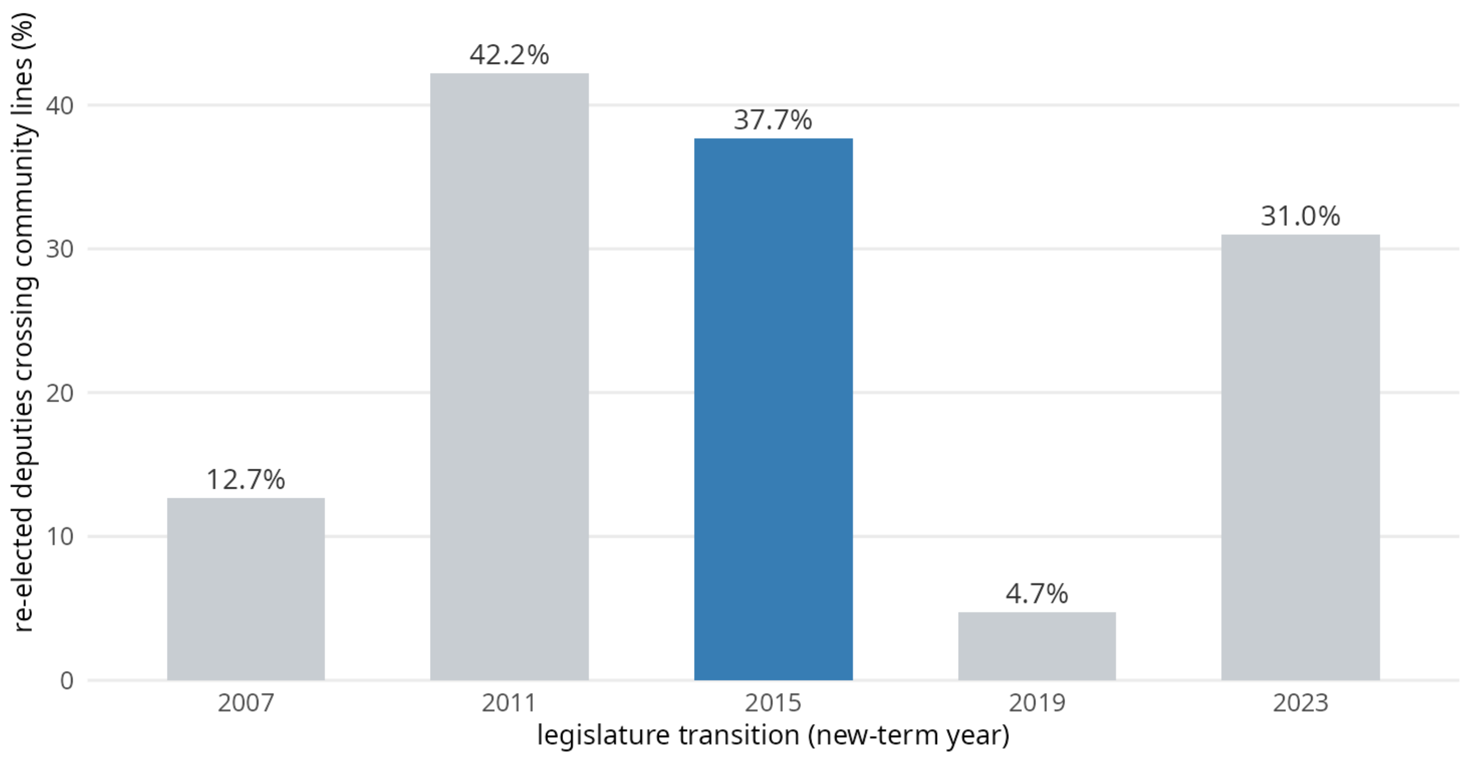
Newman–Girvan modularity, benchmarked against a permutation null: Q* is what exceeds chance.

Null-corrected modularity (Q*) over time



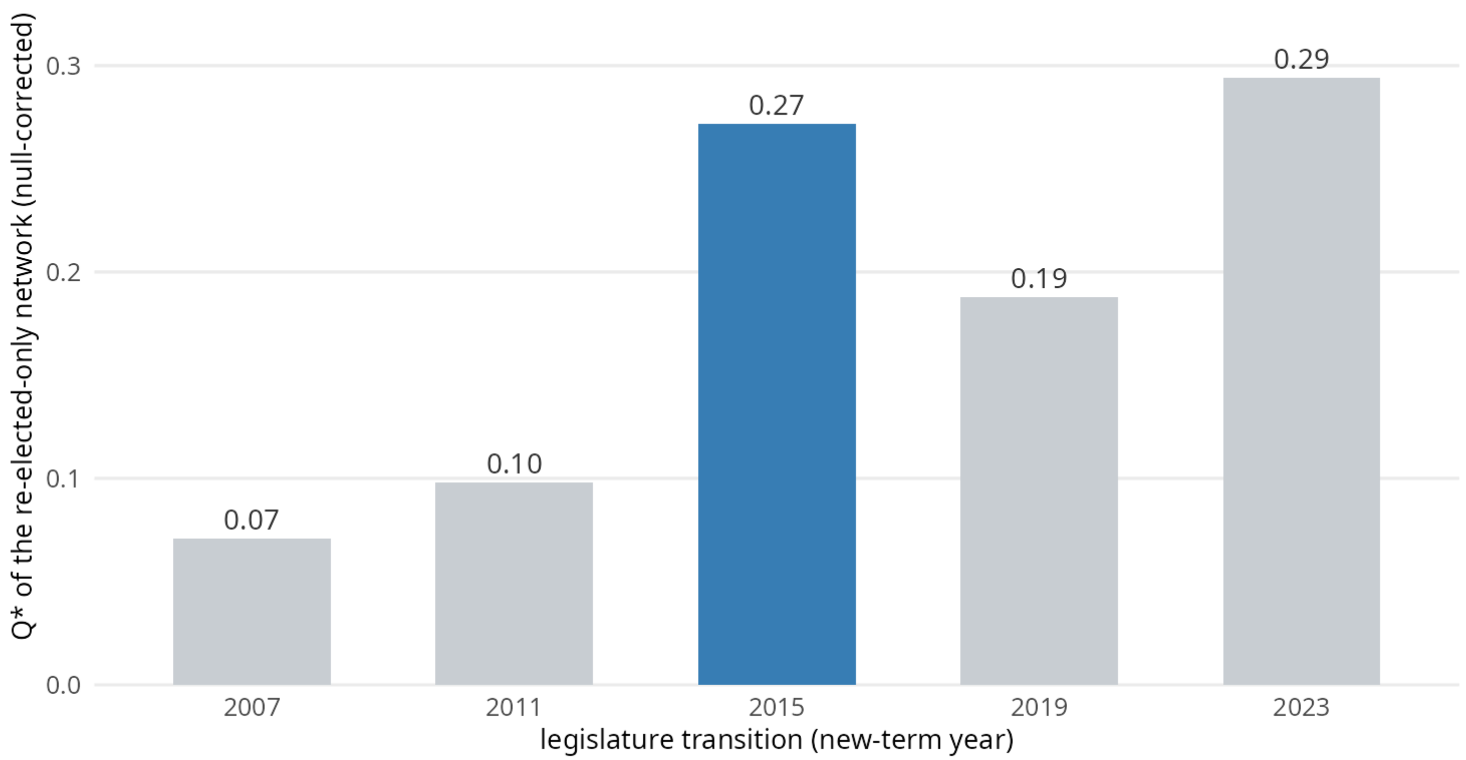
Observed Q exceeds the within-vote null, peaking in 2015–16 and rising again under Lula III.

Same people, not turnover



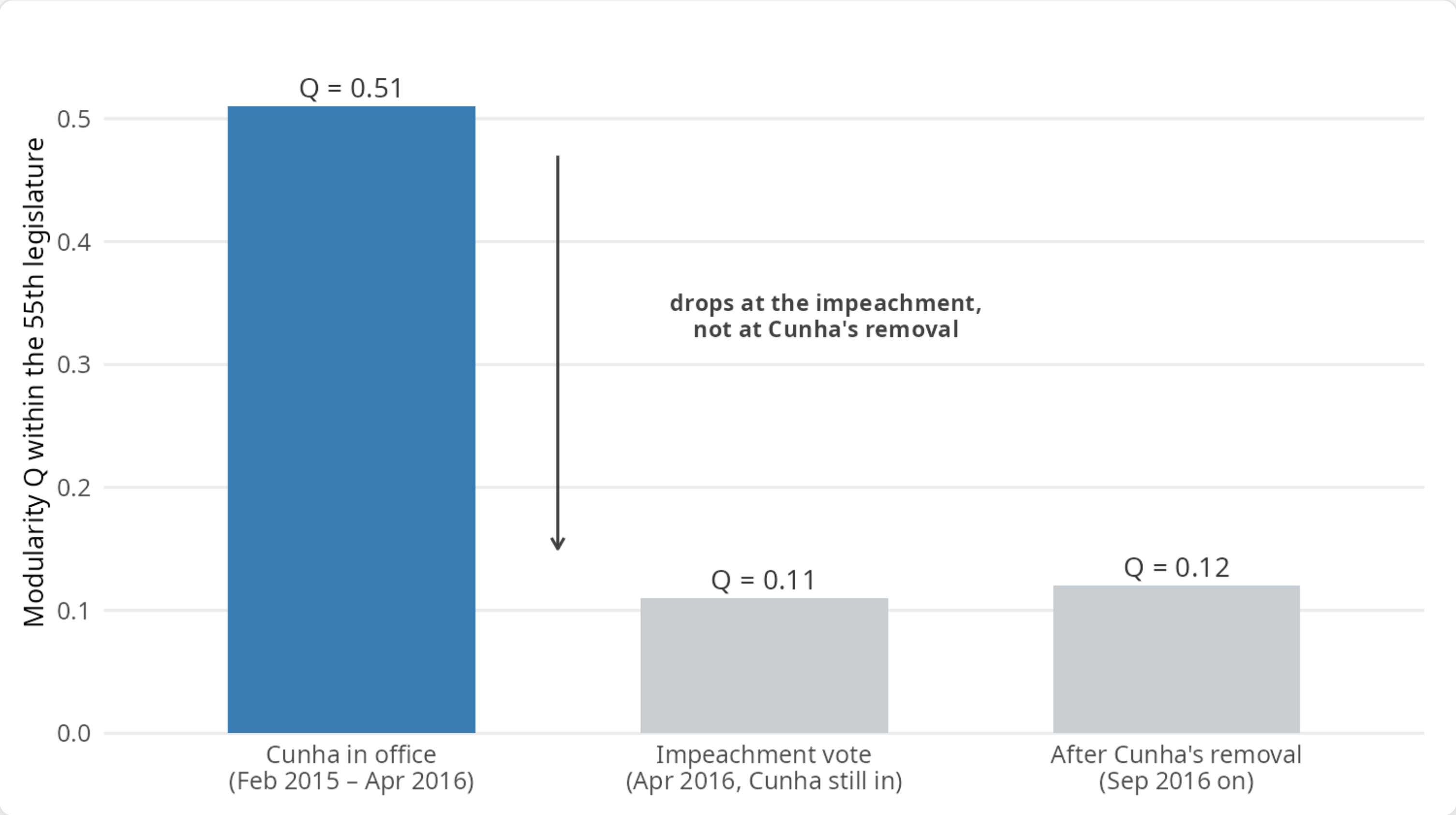
Re-elected crossing, per transition: **42%** in 2011, **37.7%** at the crisis, **4.7%** in the quiet transition.

Movement is not crystallization



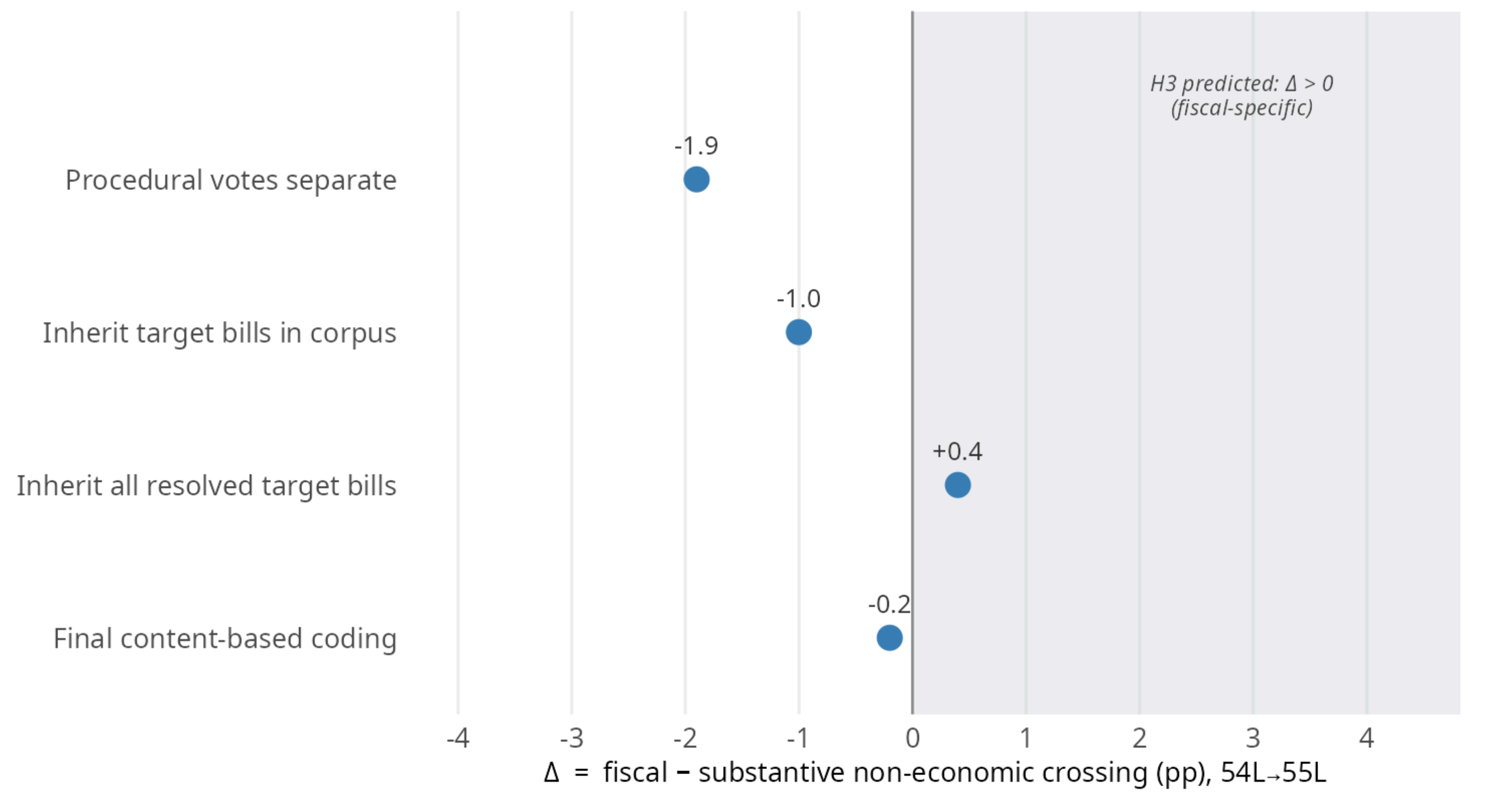
Same axis, different metric: the 2011 that moved most barely crystallizes; only **2015 and 2023** sharpen the structure.

Not Cunha's agenda: dated by the impeachment



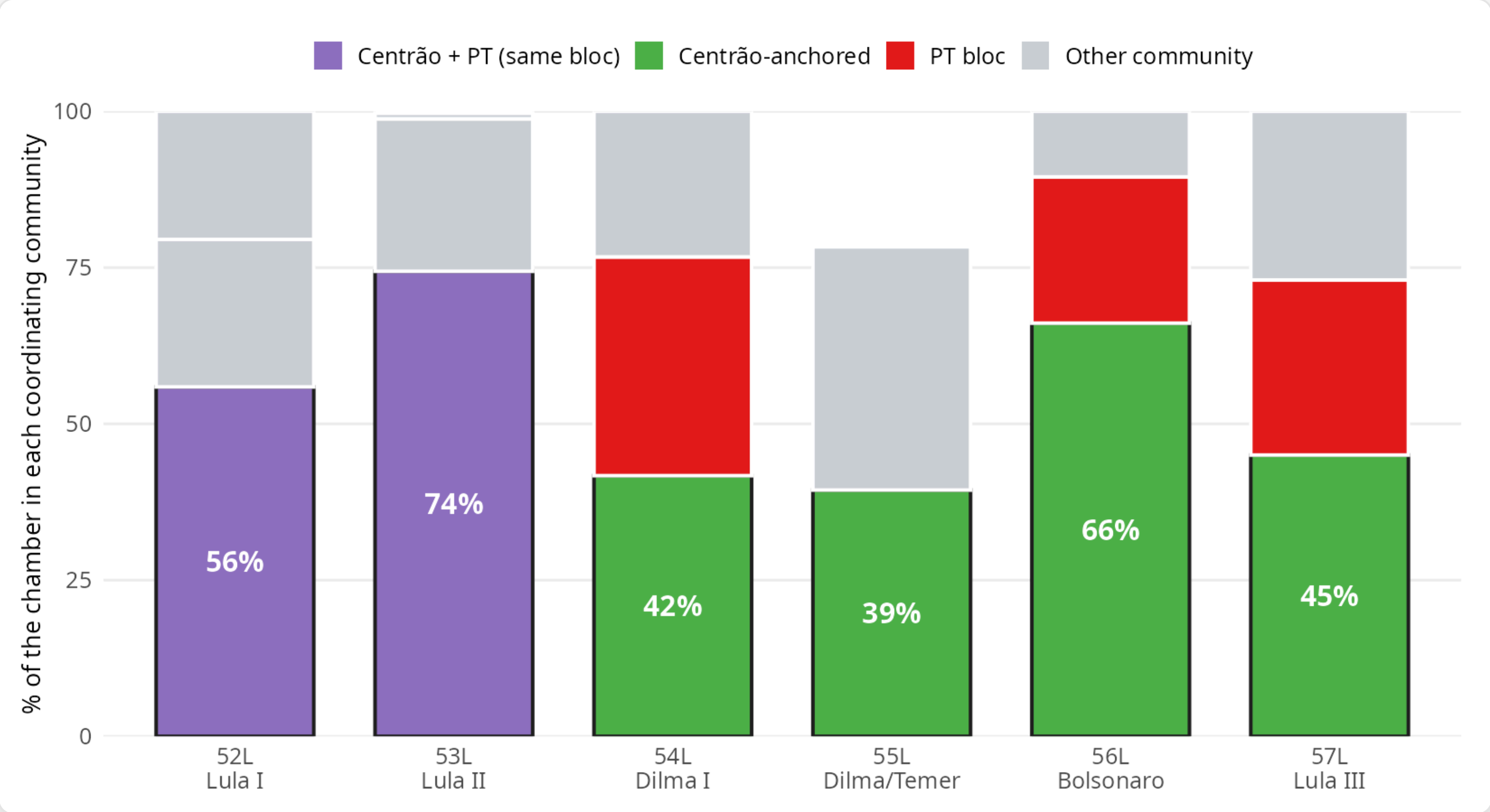
Modularity falls at the impeachment, not at Cunha's removal.

Not fiscal-specific: the fiscal contrast stays near zero



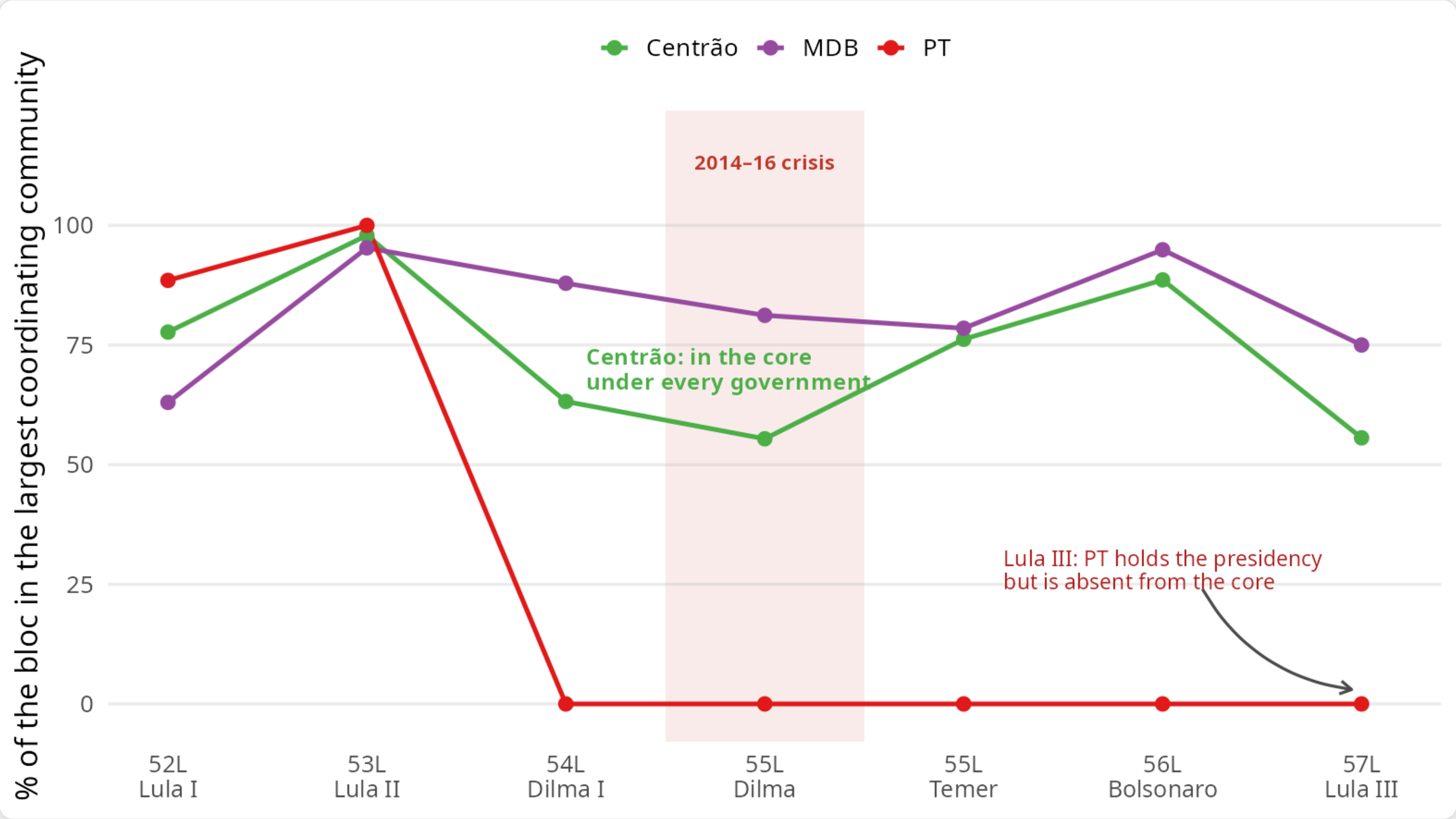
The four schemes differ only in how **procedural votes** get an issue domain. The fiscal contrast (fiscal – non-economic crossing, 54L → 55L) stays **−1.9 to +0.4 pp**: no substantively meaningful fiscal excess.

After the break, the Centrão anchors the largest community



Lula I and II: the largest bloc is the Centrão and PT together (purple). From the 2014 break the Centrão anchors it (green) and the PT drops to a smaller, separate community (red), even under Lula III.

The pivot: in the largest community under every government

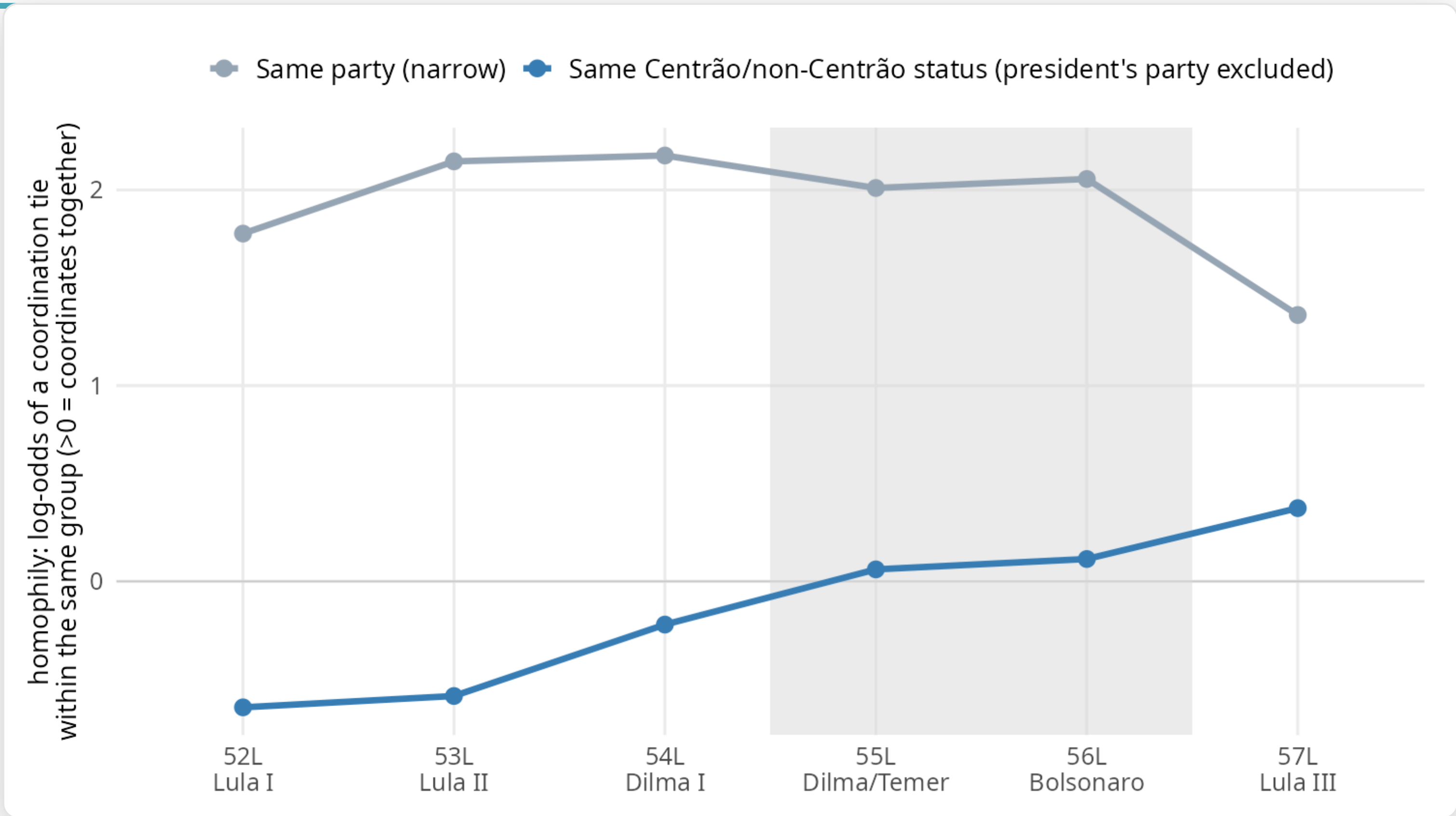


The Centrão remains in the largest community across governments; the PT does not.

Can we trust these communities?

- **Noise?** Q* sits far above a within-vote permutation null ($z > 44$).
- **Projection?** A projection-free bipartite SBM shows compatible alignment at the 56L (ARI 0.23), weakening the projection-artifact interpretation.
- **Labels?** Vote-only communities increasingly align with an independently coded Centrão (ARI 0 → 0.39).

Centrão status becomes an organizing boundary after the crisis



Same-status coordination turns positive after the crisis, net of shared party and state, with the president's party excluded from the Centrão coding. Party homophily persists.

The thesis, and what it means

- **The pragmatic center is now the recurring pivot governments build majorities through**, coordinating by **position relative to the executive**, not a fixed ideological axis.
- The Centrão is **old** (1987); what changed is its **position**, *agent* → *pivot*.
- **Why (a reading, not a tested cause):** the budget shifted to Congress (**mandatory amendments**, then the **secret budget** and **agenda control**), so the tool that once disciplined the pivot now lets it **set the terms** (*governo congressual*).
- **What it means:** governability **sold, not represented**; the cost is **accountability**, not gridlock.

Contribution, and what the case teaches

- **Measurement:** recover the Centrão's permanence from the votes, its position and persistence, where the literature had narrative.
- **Design:** with near-collinear shocks, weigh the rival explanations rather than estimate one effect.
- **A portable template:** turn a narrative prediction, that institutional crisis reshapes the legislature, into a measurable, falsifiable object, and narrow the viable mechanisms.
- **A way of looking, not a new claim:** the permanent-pivot type is old (Israel, post-war Italy).

Thank you

Questions welcome.

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